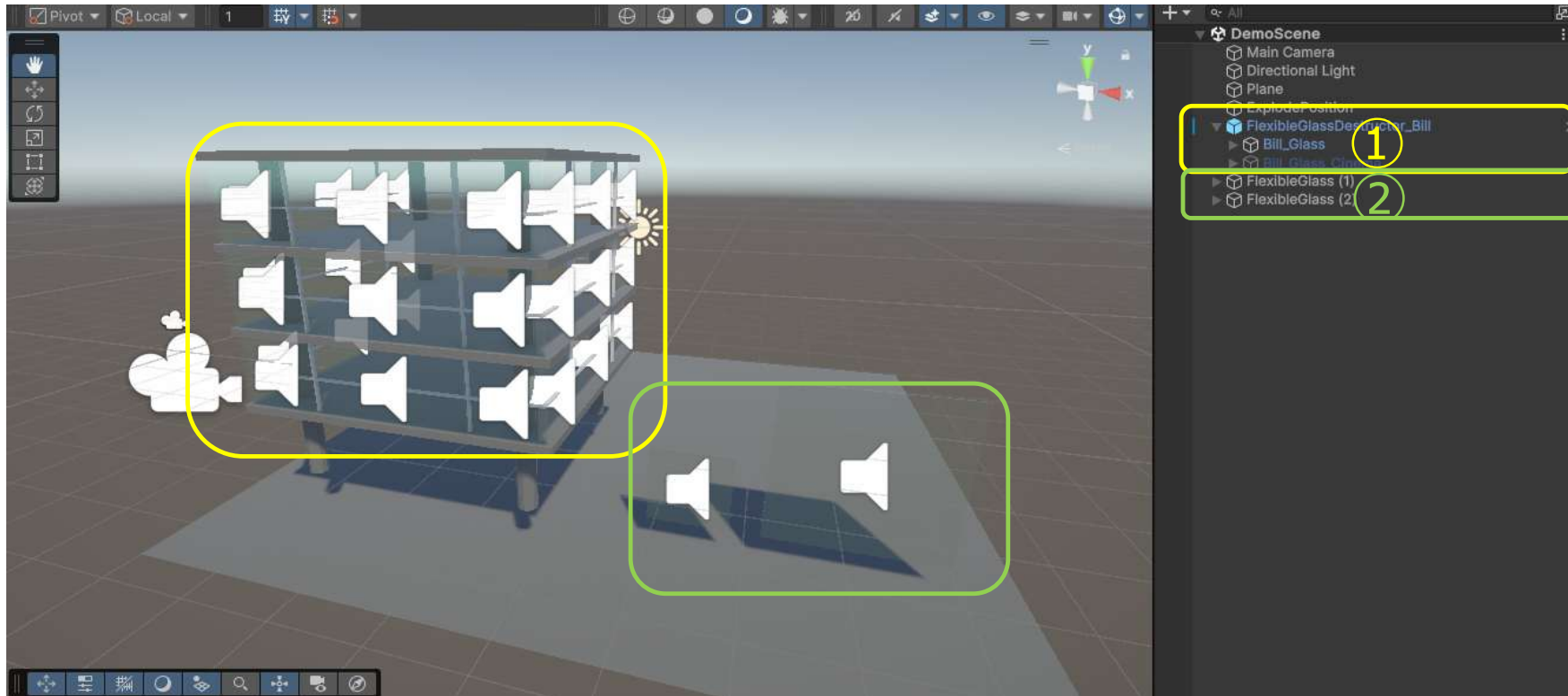


How To Use FlexibleGlassDestructor

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1. DemoScene



①Glass installed in a building

②Basic Glass

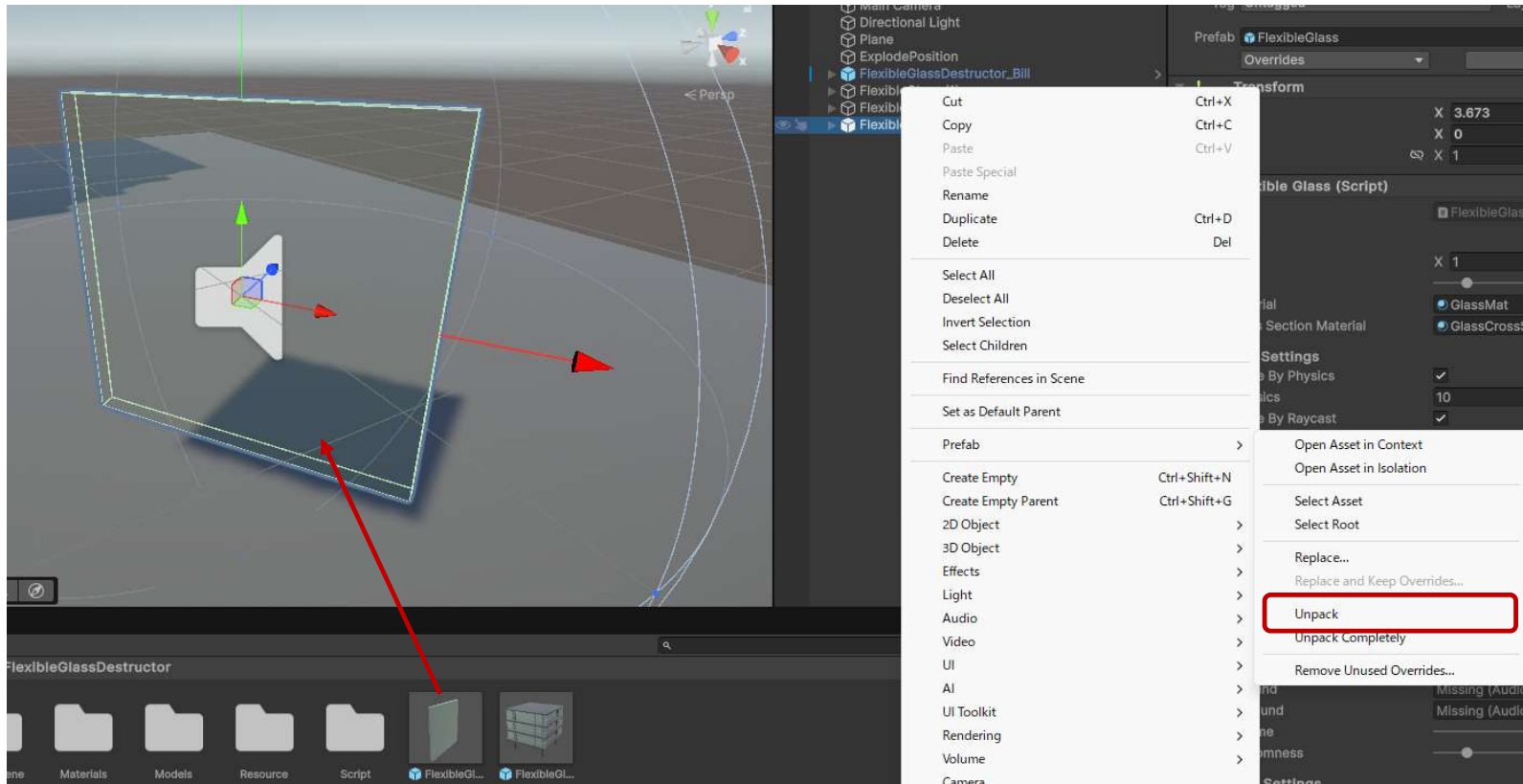
The placement may change due to updates or changes.

2. Quick Start (3 Steps)

Step 1: Setup

Drag the prefab into your scene and Unpack it.

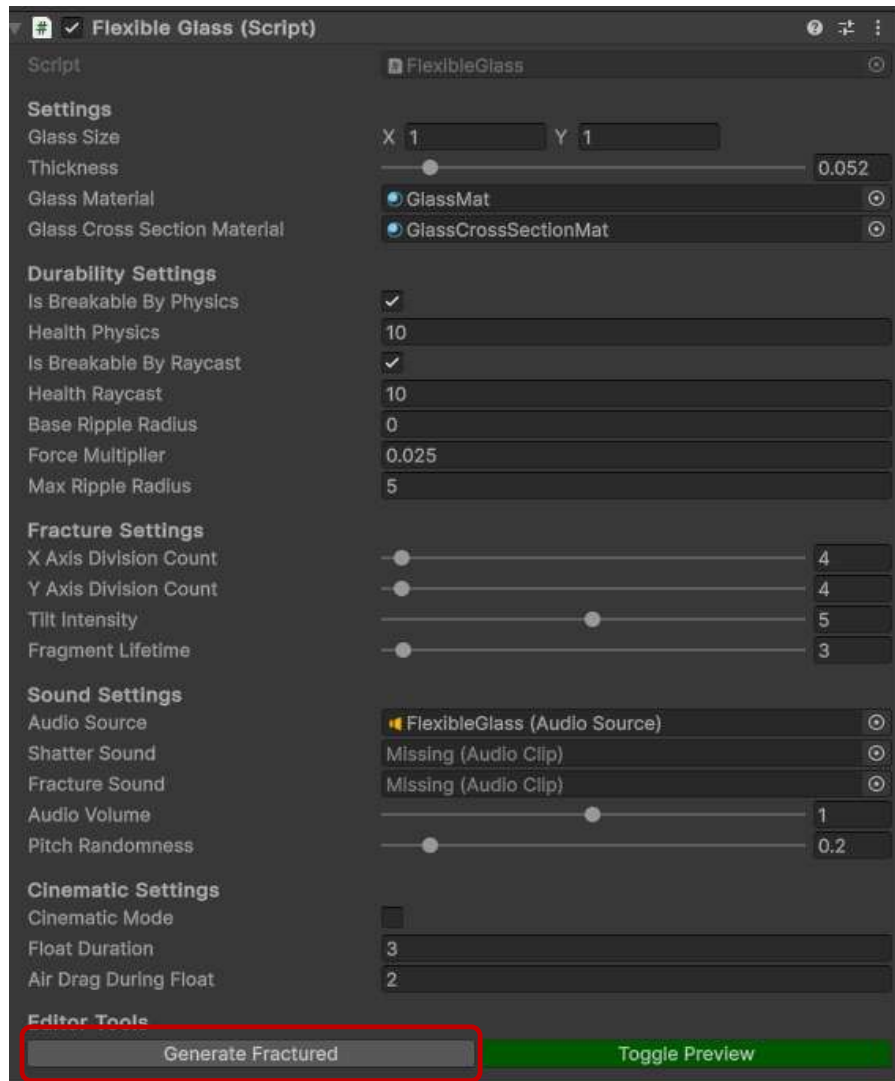
- Place the Flexible Glass prefab in the Hierarchy.
- Right-click and select "Unpack Prefab" to enable custom generation.



Step 2: Configuration & Generation

Adjust the size and generate fragments.

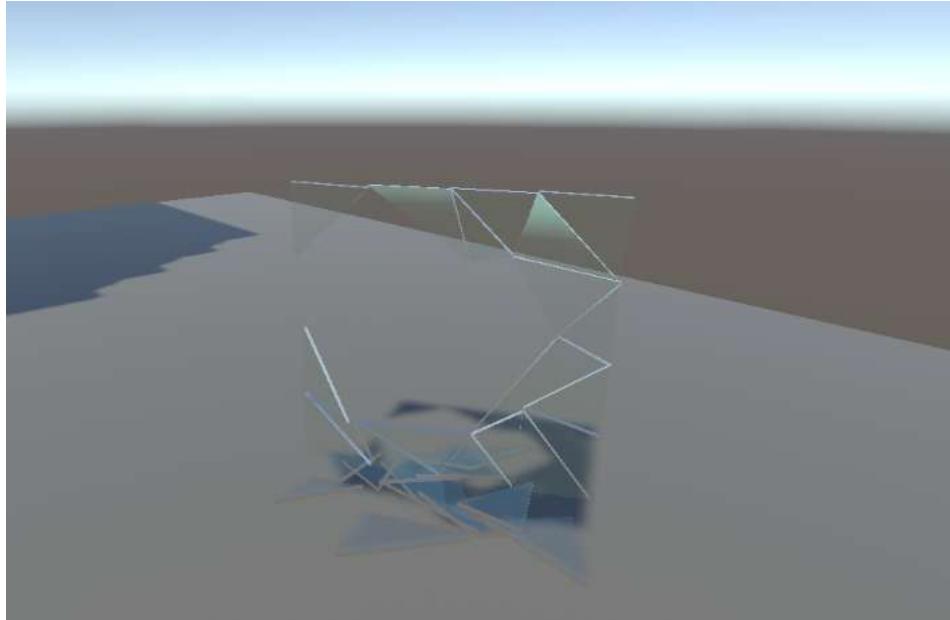
- Use the Scene View handles or Inspector values to set the glass dimensions.
- Click the "Generate Fragments" button to bake the fracture pattern.



Step 3: Action

Trigger destruction via Collision or Raycast.

- Collision: Impact the glass with a Rigidbody object.
- Raycast: Call the destruction method from your shooting script.
- Enter Play Mode to see the shattering in action!



If you intend to use this outside of the demo scene, you will need a destruction script. The sample source code is implemented in 'DemoSceneManager'. Normally, please refer to 'IGlassBreakable'.

3. Parameters

[Settings]

Glass Size	-	Width and height dimensions of the glass pane
Thickness	0.001 - 0.5	Thickness of the glass pane
Glass Material	-	Material applied to the main glass surfaces
Glass Cross Section Material	-	Material applied to the fractured cross-sections

[Durability Settings]

Is Breakable By Physics	-	Toggle for enabling/disabling physics-based destruction
Health Physics	-	Maximum durability against physical collisions
Is Breakable By Raycast	-	Toggle for enabling/disabling weapon-based destruction
Health Raycast	-	Maximum durability against raycast-based attacks
Base Ripple Radius	- (Default: 0.2)	Minimum fracture radius
Force Multiplier	- (Default: 0.05)	How much the impact force scales the radius
Max Ripple Radius	- (Default: 2.0)	Maximum allowed radius to prevent excessive spread

[Fracture Settings]

X Axis Division Count	1 - 100	Subdivisions along the horizontal axis
Y Axis Division Count	1 - 100	Subdivisions along the vertical axis
Iilt Intensity	0 - 10	Multiplier for the variation in fragment angles
Fragment Lifetime	1.0 - 60.0	Time (seconds) from when the fragment awakens until it disappears

[Sound Settings]

Audio Source	-	Component for playing sound effects
Shatter Sound	-	Audio clip for complete glass destruction
Fracture Sound	-	Audio clip for glass fracture/cracking
Audio Volume	- (Default: 1.0)	Volume of the sound effects
Pitch Randomness	- (Default: 0.2)	Amount of random variation in sound pitch

[Cinematic Settings]

Cinematic Mode	-	Toggle for cinematic visual effects
Float Duration	-	How long the fragments remain floating in the air
Air Drag DuringFloat	- (Default: 2.0)	Air resistance applied while floating to slow down movement

4. Notes / Requirements

Technical Notes & Considerations

Component Dependencies

Rigidbody Requirement:

A Rigidbody component is required.

Upon initialization, it will automatically be set to `isKinematic = true` to ensure stability.

Auto-AudioSource:

If no AudioSource is present on the object,
one will be automatically generated and attached at runtime/initialization.

Material & Rendering

Dual-Submesh Architecture:

The generated fragments consist of two submeshes (surface/back and cross-section).

Two materials must be assigned to ensure correct rendering of the glass and its edges.

Editor Behavior

Real-time Preview:

Thanks to the [ExecuteInEditMode] attribute, changes to dimensions and materials are reflected in the Scene view instantly, even when not in Play Mode.

Intentional Limitations & Safety Measures

Performance Guard:

To prevent CPU overhead, the Division Count is capped at 100 via the Inspector slider.

Collision Optimization:

To avoid physics instability, inter-fragment collisions are ignored by default.

Mesh Integrity:

Internal vertex jitter is limited to a maximum of 0.4 to prevent mesh artifacts or overlapping vertices.